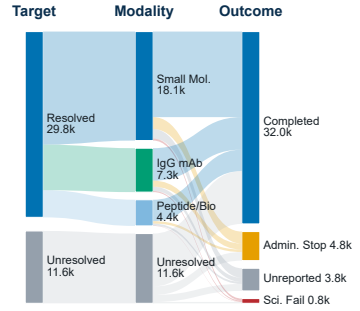
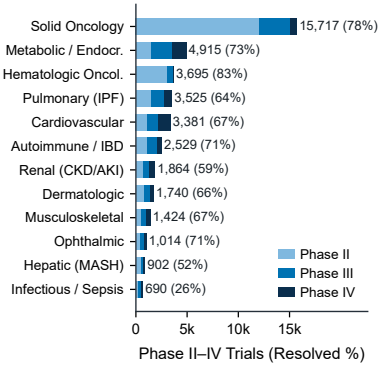
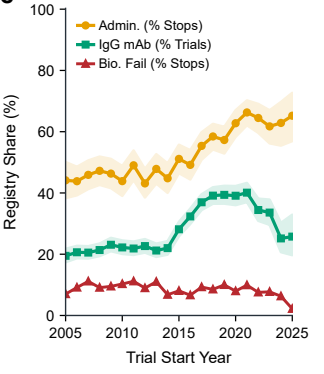
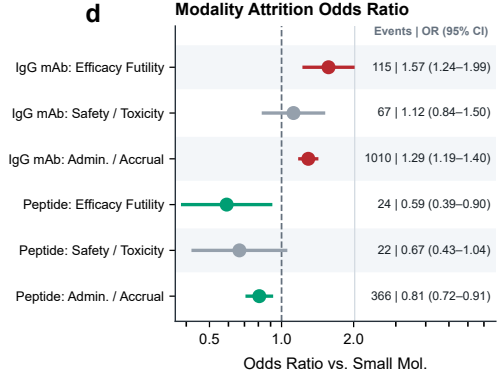
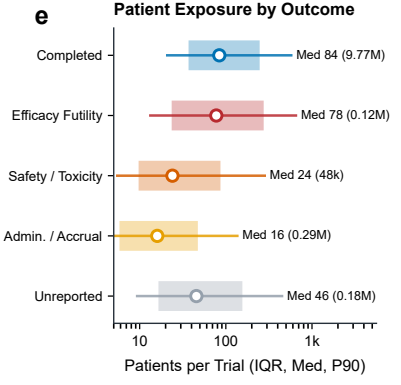
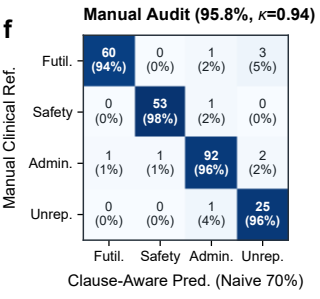

a
Trial Deconvolution (n = 41,396)
Target
Modality
Outcome
Resolved
29.8k
Unresolved
11.6k
Small Mol.
18.1k
IgG mAb
7.3k
Peptide/Bio
4.4k
Unresolved
11.6k
Completed
32.0k
Admin. Stop 4.8k
Unreported 3.8k
Sci. Fail 0.8k


b
12 Peripheral Organ Axes
Solid Oncology
Metabolic / Endocr.
Hematologic Oncol.
Pulmonary (IPF)
Cardiovascular
Autoimmune / IBD
Renal (CKD/AKI)
Dermatologic
Musculoskeletal
Ophthalmic
Hepatic (MASH)
Infectious / Sepsis
15,717 (78%)
4,915 (73%)
3,695 (83%)
3,525 (64%)
3,381 (67%)
2,529 (71%)
1,864 (59%)
1,740 (66%)
1,424 (67%)
1,014 (71%)
902 (52%)
690 (26%)
Phase II
Phase III
Phase IV
0
5k
10k
15k
Phase II–IV Trials (Resolved %)


c
20-Year Modality & Stoppage Drift
Admin. (% Stops)
IgG mAb (% Trials)
Bio. Fail (% Stops)
Registry Share (%)
100
80
60
40
20
0
2005
2010
2015
2020
2025
Trial Start Year


d
Modality Attrition Odds Ratio
Events | OR (95% CI)
IgG mAb: Efficacy Futility
IgG mAb: Safety / Toxicity
IgG mAb: Admin. / Accrual
Peptide: Efficacy Futility
Peptide: Safety / Toxicity
Peptide: Admin. / Accrual
115 | 1.57 (1.24–1.99)
67 | 1.12 (0.84–1.50)
1010 | 1.29 (1.19–1.40)
24 | 0.59 (0.39–0.90)
22 | 0.67 (0.43–1.04)
366 | 0.81 (0.72–0.91)
0.5
1.0
2.0
Odds Ratio vs. Small Mol.


e
Patient Exposure by Outcome
Completed
Efficacy Futility
Safety / Toxicity
Admin. / Accrual
Unreported
Med 84 (9.77M)
Med 78 (0.12M)
Med 24 (48k)
Med 16 (0.29M)
Med 46 (0.18M)
10
100
1k
Patients per Trial (IQR, Med, P90)


f
Manual Audit (95.8%, κ=0.94)
Manual Clinical Ref.
Futil.
Safety
Admin.
Unrep.
60 (94%)
0 (0%)
1 (2%)
3 (5%)
0 (0%)
53 (98%)
1 (2%)
0 (0%)
1 (1%)
1 (1%)
92 (96%)
2 (2%)
0 (0%)
0 (0%)
1 (4%)
25 (96%)
Futil.
Safety
Admin.
Unrep.
Clause-Aware Pred. (Naive 70%)